

3.6.5 Electromagnetic waves

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 9			
Electromagnetic waves are transverse waves that transfer energy from the source of the waves to an absorber.	Students will not be required to memorise the values of the wavelength or frequency of these waves.	6.6.2.1	4.1.4.3
Electromagnetic waves form a continuous spectrum and all types of electromagnetic wave travel at the same velocity through a vacuum (space) or air. The waves that form the electromagnetic spectrum are grouped in terms of their wavelength and their frequency. Going from long to short wavelength (or from low to high frequency) the groups are: • radio • microwave • infrared • visible light (red to violet) • ultraviolet • X-rays • gamma rays. Ultraviolet waves, X-rays and gamma rays can have hazardous effects on human body tissue. The effects depend on the type of radiation and the size of the dose.	Suggested activity for TDA Testing visual acuity in different colours of light.	6.6.2.1	4.1.4.3
Outcome 10			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Electromagnetic waves have many practical applications, eg: • radio waves – television and radio (including Bluetooth) • microwaves – satellite communications, cooking food • infrared – electrical heaters, cooking food, infrared cameras • visible light – fibre optic communications • ultraviolet – energy efficient lamps, sun tanning • X-rays – medical imaging and treatments • gamma rays – for sterilising. Students should be able to give brief explanations of why each type of electromagnetic wave is suitable for the practical application.	Suggested activity for TDA Investigate the shielding of a mobile phone or remote control device. Suggested activity for TDA Investigate the range over which a Bluetooth device is effective.	6.6.2.4	4.1.4.3